

Dropped Foot, Hidden Culprit: *Electrodiagnostic Clarity in a Case of Occupational Peroneal Neuropathy Masquerading as Lumbar Radiculopathy*

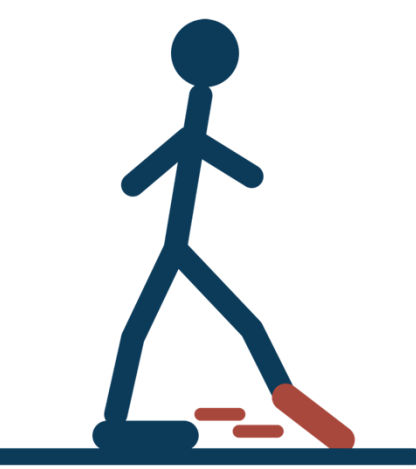
Max Thrush Hukill AB¹, Dasol Lee BA¹, Anne Kuwabara MD¹

¹Kaiser Permanente Bernard J. Tyson School of Medicine, Pasadena, California

Clinical Presentation

HISTORY OF PRESENT ILLNESS

54-year-old man · 3 weeks of progressive left foot drop



Toe dragging; tripping, altered gait



Numbness over dorsal left foot



Intermittent low back pain



HVAC & construction technician

EXAMINATION & LABS

- Unremarkable vitals
- Relevant physical exam findings below:

Sensation	Findings
L5 (webpace between 1 st /2 nd toe)	↓ left side
S1 (lateral malleolus)	↓ left side
All other dermatomes in b/l LE	Intact

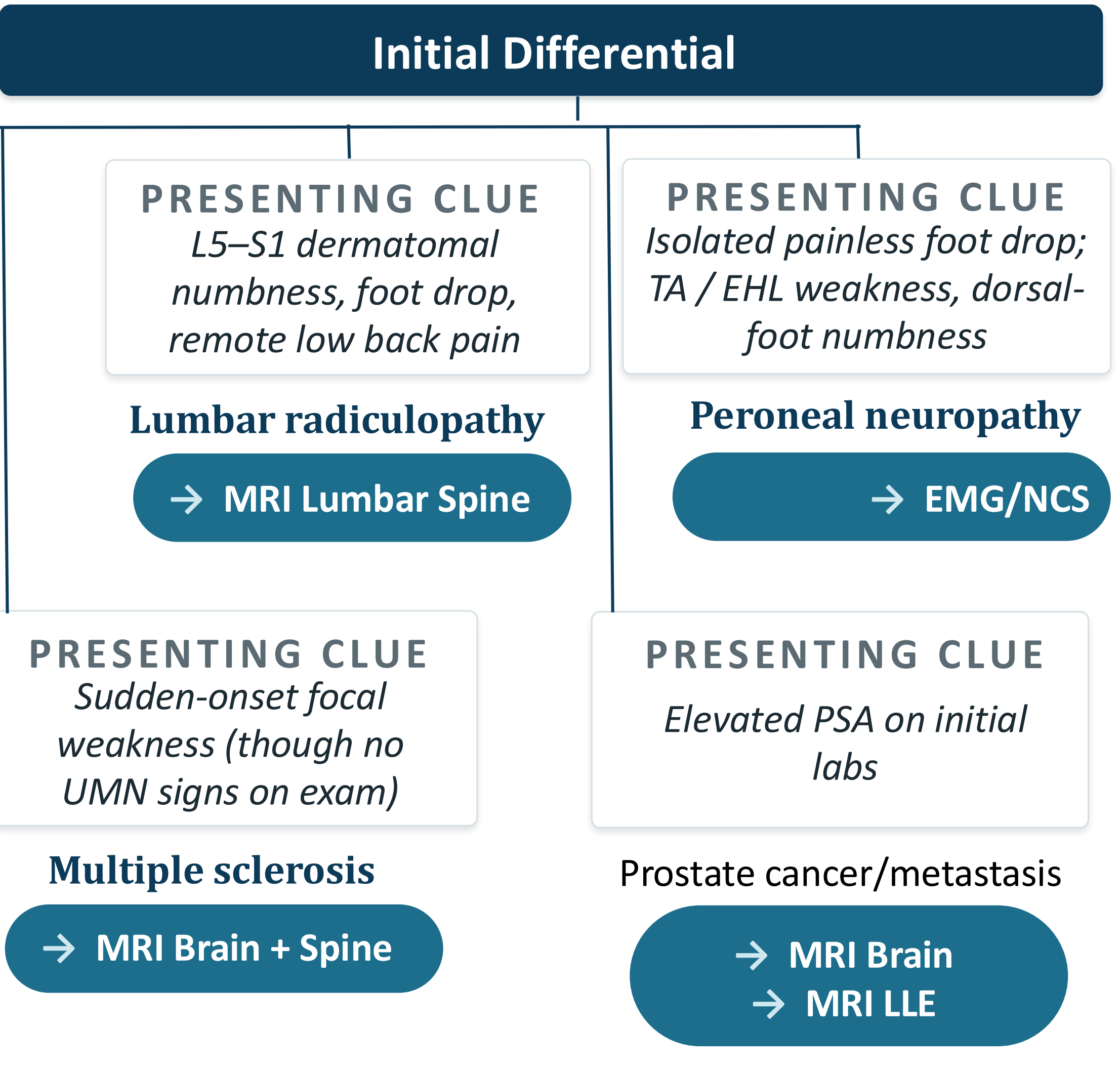
Strength	Grade	Test	Value
Tibialis Anterior	0/5	H _{A1C}	5.5%
Extensor Hallicus Longus	0/5	WBC	7.6
All other	5/5	HBG	14.8

Reflexes	Grade
Biceps	0
Brachioradialis	0
Triceps	0
Patellar	0
Achilles	0
Hoffman	(-)
Ankle clonus	(-)

- Other significant findings:
- Negative Spurling
 - Left foot drop

Localizing wrinkle: dermatomal numbness + diffuse areflexia blurred the picture → possible root lesion?

Initial Reasoning



Resolution:

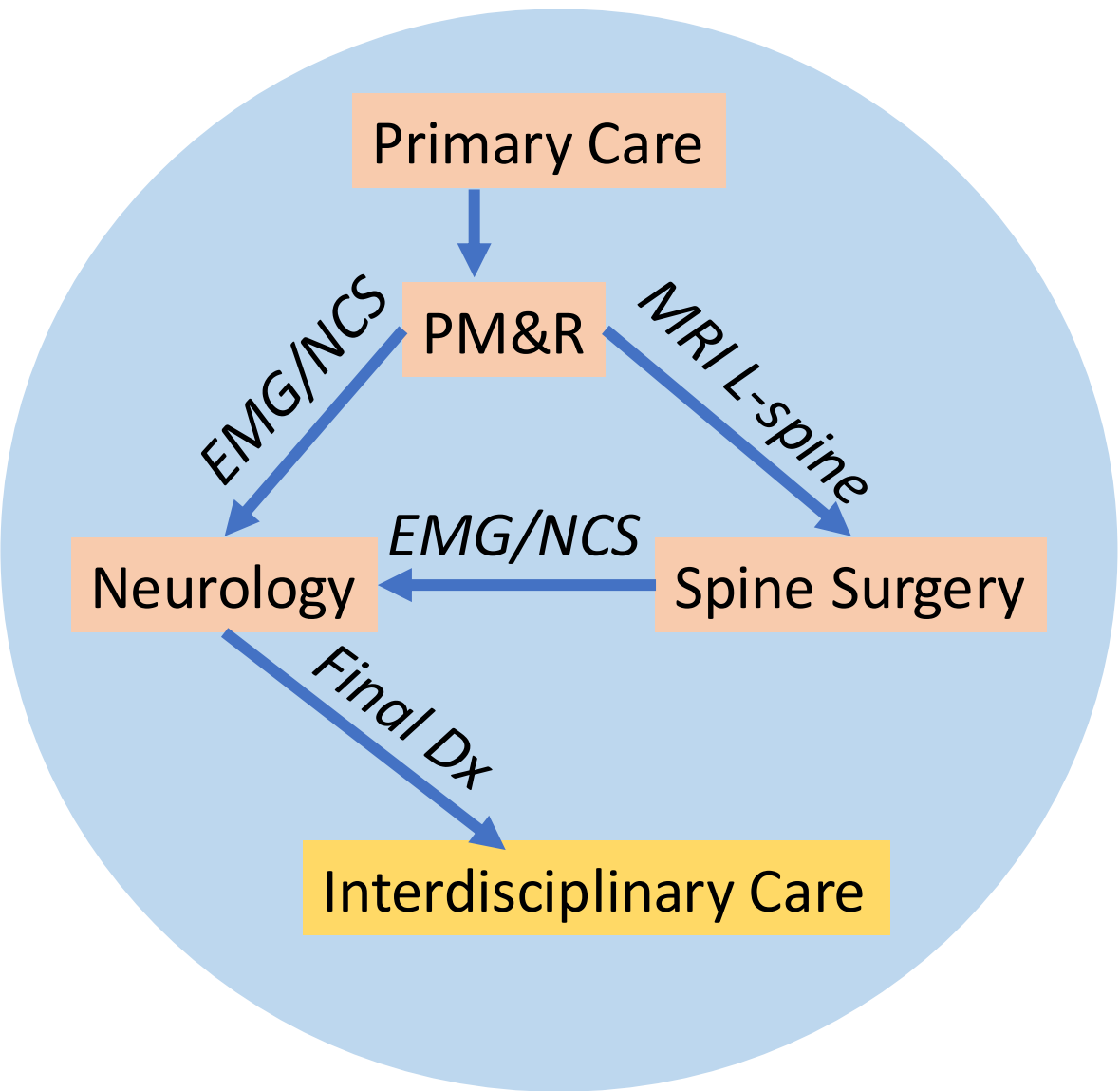
Initial differential	Verdict
Lumbar radiculopathy <i>exiting left L5 and S1</i>	RED HERRING
Peroneal neuropathy <i>at the fibular neck</i>	CONFIRMED
Multiple sclerosis	RULED OUT
Prostate cancer / metastasis	RULED OUT

Treatment: physical therapy, lifestyle modification, offloading orthosis

While MRI suggested L5–S1 radiculopathy, EDX localized the foot drop to peroneal neuropathy.

Remember to:

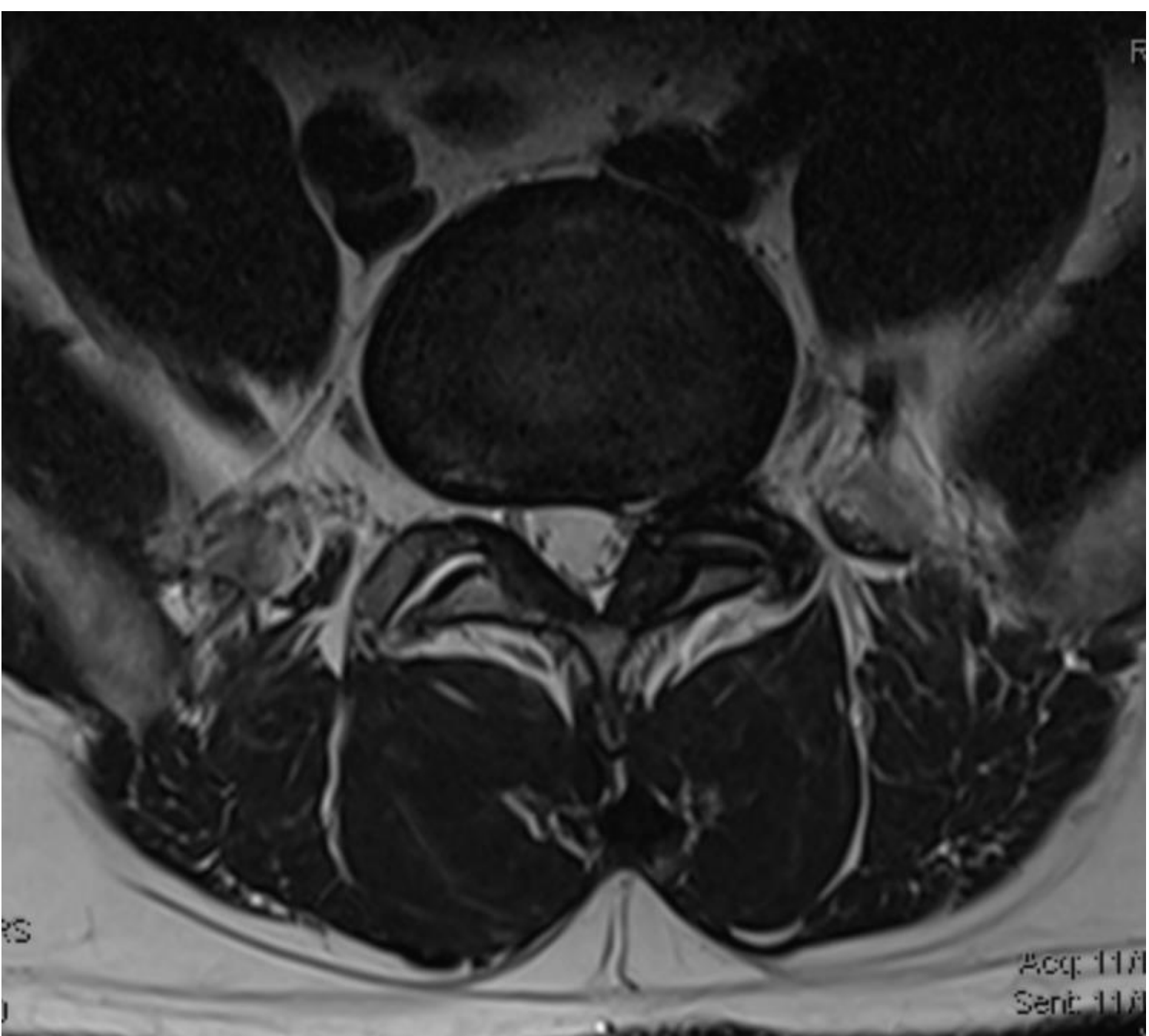
- Integrate clinical history with electrodiagnostic testing
- Lean on interdisciplinary thinking



The Red Herring: MRI



Sagittal T2 view: left foraminal stenosis at L5-S1 potentially affecting L5 nerve root



Axial T2 view: at L5-S1, left foraminal stenosis potentially affecting L5 nerve root

A COMPELLING CLUE:

Foraminal stenosis + dermatomal numbness + foot drop all seemed to fit a left L5 radiculopathy.

A left L5–S1 hemilaminotomy with microdiscectomy was on the table.

The Answer: EMG / NCS

Nerve Conduction Studies
Anti Sensory Summary Table

Stim Site	NR	Peak (ms)	Norm Peak (ms)	O-P* Amp (µV)	Norm O-P Amp	Site1	Site2	Delta-P (ms)
Left Sup Peron Anti Sensory (Ant Lat Mall)								
14 cm	NR		<4.5		>5.0	14 cm	Ant Lat Mall	
Site 2 NR								
Left Sural Anti Sensory (Lat Mall)								
Calf 14cm		3.6	<4.5	13.4	>5	Calf 14cm	Lat Mall	3.6

→ Absent superficial peroneal SNAP with preserved sural SNAP favors a peripheral peroneal neuropathy over an L5 radiculopathy

EMG

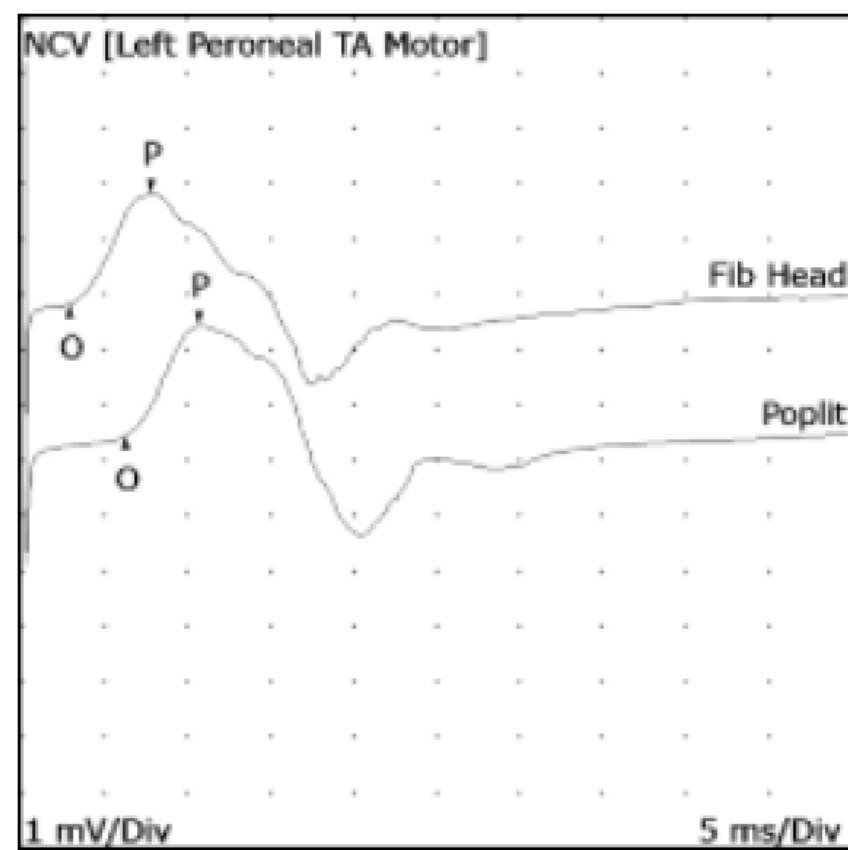
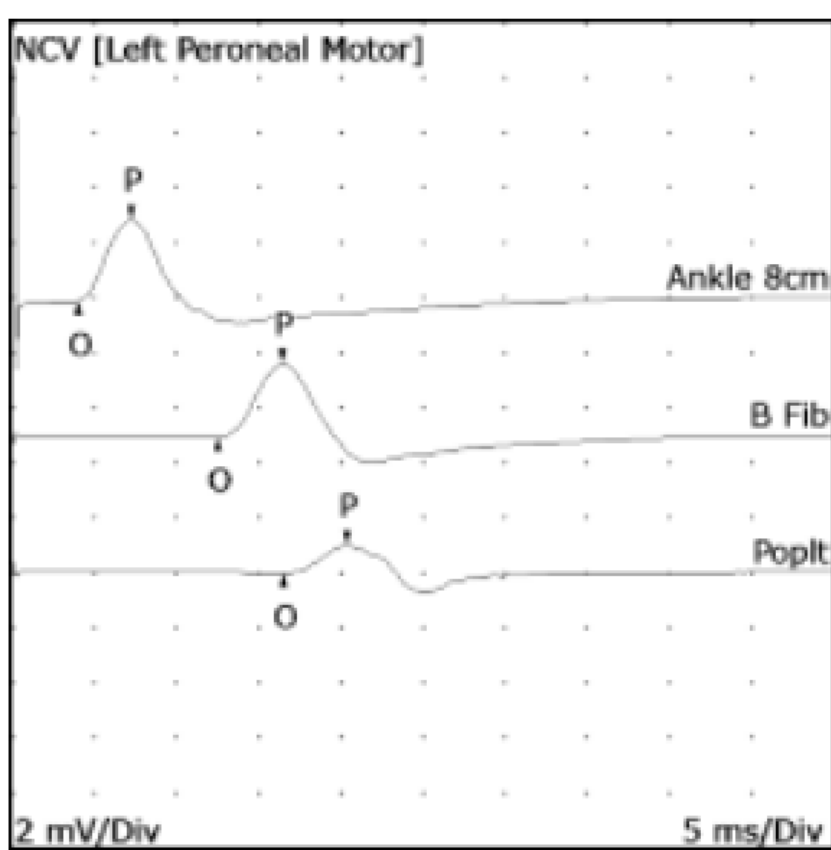
Side	Muscle	Nerve	Root	Fibs/PSW	Amp	Dur	Poly	Recrt	Comment
Left	AntTibialis	Dp Br Peron	L4-5	1+	Nml	Nml	0	Nml	
Left	MedGastroc	Tibial	S1-2	Nml	Nml	Nml	0	Nml	
Left	BicepsFemS	Sciatic	L5-S1	Nml	Nml	Nml	0	Nml	

→ Fibrillations in tibialis anterior; short-head biceps femoris & gastrocnemius spared so denervation confined to the peroneal distribution

Motor Summary Table

Stim Site	NR	Onset (ms)	Norm Onset (ms)	O-P Amp (mV)	Norm O-P Amp	Site1	Site2	Delta-0 (ms)	Dist (cm)	Vel (m/s)	Norm Vel (m/s)
Left Peroneal Motor (Ext Dig Brev)											
Ankle 8cm		4.1	<6.5	3.0	>2.0	Ankle 8cm	B Fib	8.4	40.0	48	>38
B Fib		12.5		2.7		B Fib	Poplt	4.1	6.0	15	
Poplt		16.6		1.0							
Left Peroneal TA Motor (Tib Ant)											
Fib Head		2.9	<6.7	2.0	>3	Fib Head	Poplt	3.4	6.0	18	>38
Poplt		6.3		2.0							
Left Tibial Motor (Abd Hall Brev)											
Ankle 9cm		4.3	<6.6	10.0	>2	Ankle 9cm	Knee	10.8	42.0	39	>38
Knee		15.1		8.0							

→ Focal slowing (≈18 m/s) with amplitude drop across the fibular neck, Tibial CMAP normal



→ Focal slowing with conduction block at the fibular neck

IMPRESSION

Acute left common peroneal neuropathy with conduction block at the fibular neck.